11.2.3 Nano in Nature

Nanoscale structures existed in nature long before scientists began studying them in laboratories.

A few examples



Nano in Nature

Object

A single strand of DNA, the building block of all living things, is about three nanometers wide.

Mimic in laboratories

Manipulation of colours by adjusting the size of nano particles with which the materials are made.



Nano in Nature

Object

The scales on the wings of a morpho butterfly contain nanostructures that change the way light waves interact with each other, giving the wings brilliant metallic blue and green hues.

Object

Peacock feathers get their iridescent coloration from light interacting with 2 dimensional photonic crystal structures just tens of nanometers thick.

Nano in Nature



Mimic in laboratories

Similar nano structures are made in lab to glow in different colors.

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